

change in concavity corresponds to the initial acceleration for $x < \pi/2$, followed by the deceleration toward $x = \pi$.

2. The same reasoning applies to any initial condition x_0 . Figure 2.1.1 shows that if $\dot{x} > 0$ initially, the particle heads to the right and asymptotically approaches the nearest stable fixed point. Similarly, if $\dot{x} < 0$ initially, the particle approaches the nearest stable fixed point to its left. If $\dot{x} = 0$, then x remains constant. The qualitative form of the solution for any initial condition is sketched in Figure 2.1.3.

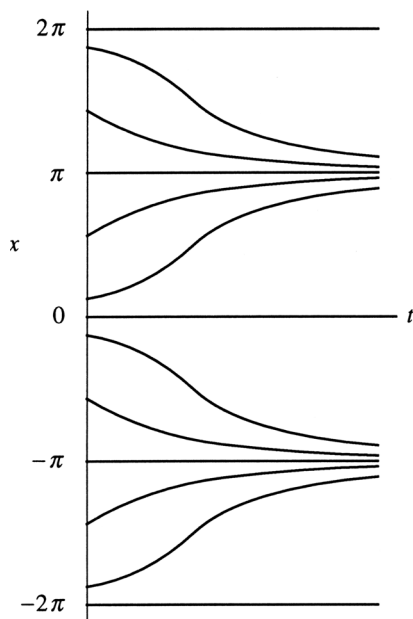


Figure 2.1.3

In all honesty, we should admit that a picture can't tell us certain *quantitative* things: for instance, we don't know the time at which the speed $|\dot{x}|$ is greatest. But in many cases *qualitative* information is what we care about, and then pictures are fine.

2.2 Fixed Points and Stability

The ideas developed in the last section can be extended to any one-dimensional system $\dot{x} = f(x)$. We just need to draw the graph of $f(x)$ and then use it to sketch the vector field on the real line (the x -axis in Figure 2.2.1).

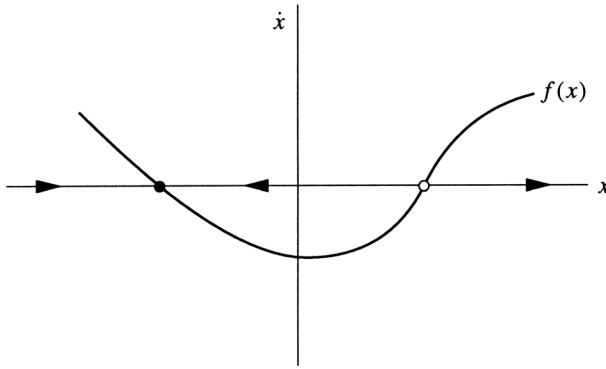


Figure 2.2.1

As before, we imagine that a fluid is flowing along the real line with a local velocity $f(x)$. This imaginary fluid is called the phase fluid, and the real line is the phase space. The flow is to the right where $f(x) > 0$ and to the left where $f(x) < 0$. To find the solution to $\dot{x} = f(x)$ starting from an arbitrary initial condition x_0 , we place an imaginary particle (known as a **phase point**) at x_0 and watch how it is carried along by the flow. As time goes on, the phase point moves along the x -axis according to some function $x(t)$. This function is called the **trajectory** based at x_0 , and it represents the solution of the differential equation starting from the initial condition x_0 . A picture like Figure 2.2.1, which shows all the qualitatively different trajectories of the system, is called a **phase portrait**.

The appearance of the phase portrait is controlled by the fixed points x^* , defined by $f(x^*) = 0$; they correspond to stagnation points of the flow. In Figure 2.2.1, the solid black dot is a stable fixed point (the local flow is toward it) and the open dot is an unstable fixed point (the flow is away from it).

In terms of the original differential equation, fixed points represent **equilibrium** solutions (sometimes called steady, constant, or rest solutions, since if $x = x^*$ initially, then $x(t) = x^*$ for all time). An equilibrium is defined to be stable if all sufficiently small disturbances away from it damp out in time. Thus stable equilibria are represented geometrically by stable fixed points. Conversely, unstable equilibria, in which disturbances grow in time, are represented by unstable fixed points.

Example 2.2.1:

Find all the fixed points for $\dot{x} = x^2 - 1$, and classify their stability.

Solution: Here $f(x) = x^2 - 1$. To find the fixed points, we set $f(x^*) = 0$ and solve for x^* . Thus $x^* = \pm 1$. To determine stability, we plot $x^2 - 1$ and then sketch the vector field (Figure 2.2.2). The flow is to the right where $x^2 - 1 > 0$

and to the left where $x^2 - 1 < 0$. Thus $x^* = -1$ is stable, and $x^* = 1$ is unstable. ■

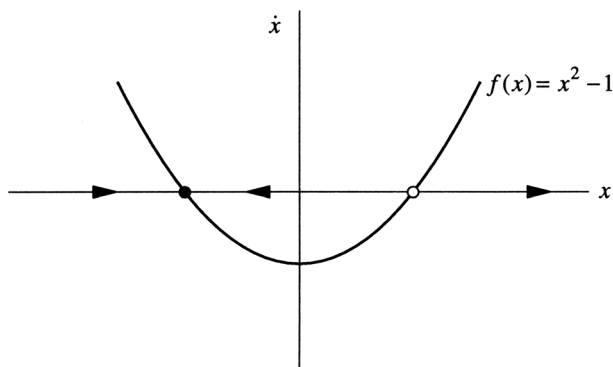


Figure 2.2.2

Note that the definition of stable equilibrium is based on *small* disturbances; certain large disturbances may fail to decay. In [Example 2.2.1](#), all small disturbances to $x^* = -1$ will decay, but a large disturbance that sends x to the right of $x = 1$ will *not* decay—in fact, the phase point will be repelled out to $+\infty$. To emphasize this aspect of stability, we sometimes say that $x^* = -1$ is *locally stable*, but not globally stable.

Example 2.2.2:

Consider the electrical circuit shown in [Figure 2.2.3](#).

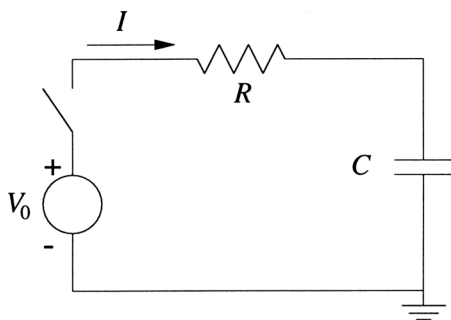


Figure 2.2.3

A resistor R and a capacitor C are in series with a battery of constant dc voltage V_0 . Suppose that the switch is closed at $t = 0$, and that there is no charge on the capacitor initially. Let $Q(t)$ denote the charge on the capacitor at time $t \geq 0$. Sketch the graph of $Q(t)$.

Solution: This type of circuit problem is probably familiar to you. It is governed by linear equations and can be solved analytically, but we prefer to illustrate the geometric approach.

First we write the circuit equations. As we go around the circuit, the total voltage drop must equal zero; hence $-V_0 + RI + Q/C = 0$, where I is the current flowing through the resistor. This current causes charge to accumulate on the capacitor at a rate $\dot{Q} = I$. Hence

$$-V_0 + R\dot{Q} + Q/C = 0$$

which gives

$$\dot{Q} = f(Q) = \frac{V_0}{R} - \frac{Q}{RC}.$$

The graph of $f(Q)$ is a straight line with a negative slope (Figure 2.2.4).

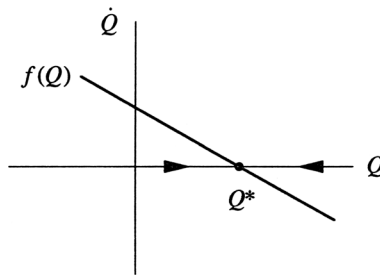


Figure 2.2.4

The corresponding vector field has a fixed point where $f(Q) = 0$, which occurs at $Q^* = CV_0$. The flow is to the right where $f(Q) > 0$ and to the left where $f(Q) < 0$. Thus the flow is always toward Q^* ; it is a *stable* fixed point. In fact, it is **globally stable**, in the sense that it is approached from *all* initial conditions.

To sketch $Q(t)$, we start a phase point at the origin of Figure 2.2.4 and imagine how it would move. The flow carries the phase point monotonically toward Q^* . Its speed $\dot{Q}$ decreases linearly as it approaches the fixed point; therefore $Q(t)$ is increasing and concave down, as shown in Figure 2.2.5. ■

Example 2.2.3:

Sketch the phase portrait corresponding to $\dot{x} = x - \cos x$, and determine the stability of all the fixed points.

Solution: One approach would be to plot the function $f(x) = x - \cos x$ and then sketch the associated vector field. This method is valid, but it requires you to figure out what the graph of $x - \cos x$ looks like.

There's an easier solution, which exploits the fact that we know how to graph $y = x$ and $y = \cos x$ *separately*. We plot both graphs on the same axes and then observe that they intersect in exactly one point (Figure 2.2.6).

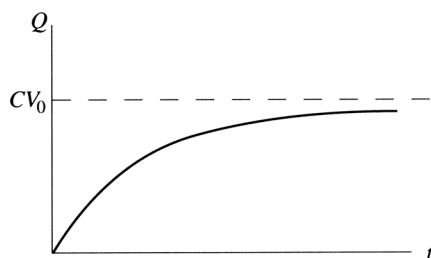


Figure 2.2.5

This intersection corresponds to a fixed point, since $x^* = \cos x^*$ and therefore $f(x^*) = 0$. Moreover, when the line lies above the cosine curve, we have $x > \cos x$ and so $\dot{x} > 0$: the flow is to the right. Similarly, the flow is to the left where the line is below the cosine curve.

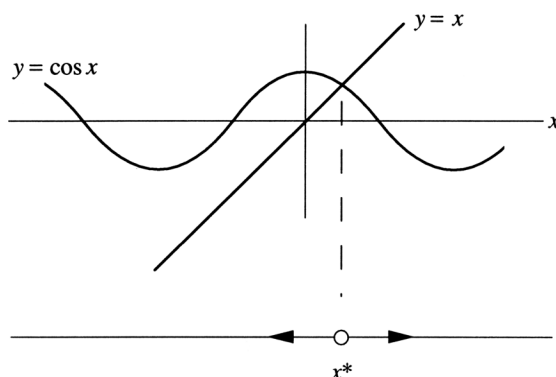


Figure 2.2.6

Hence x^* is the only fixed point, and it is unstable. Note that we can classify the stability of x^* , even though we don't have a formula for x^* itself! ■

2.3 Population Growth

The simplest model for the growth of a population of organisms is $\dot{N} = rN$, where $N(t)$ is the population at time t , and $r > 0$ is the growth rate. This model predicts exponential growth: $N(t) = N_0 e^{rt}$, where N_0 is the population at $t = 0$.

Of course such exponential growth cannot go on forever. To model the effects of overcrowding and limited resources, population biologists and

demographers often assume that the per capita growth rate $\dot{N}/N$ decreases when N becomes sufficiently large, as shown in [Figure 2.3.1](#). For small N , the growth rate equals r , just as before. However, for populations larger than a certain *carrying capacity* K , the growth rate actually becomes negative; the death rate is higher than the birth rate.

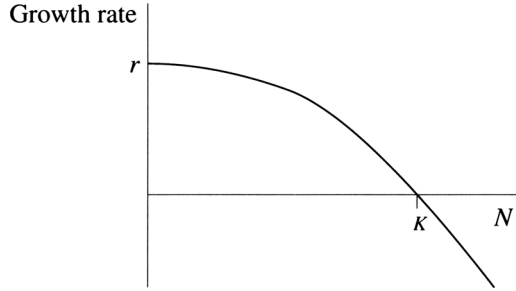


Figure 2.3.1

A mathematically convenient way to incorporate these ideas is to assume that the per capita growth rate $\dot{N}/N$ decreases *linearly* with N ([Figure 2.3.2](#)).

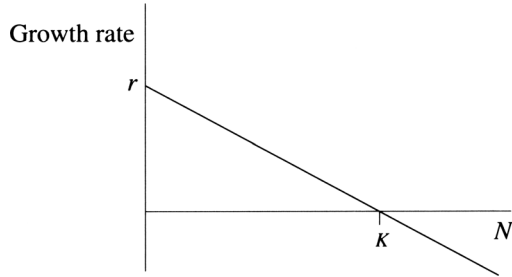


Figure 2.3.2

This leads to the *logistic equation*

$$\dot{N} = rN \left(1 - \frac{N}{K} \right),$$

first suggested to describe the growth of human populations by Verhulst in 1838. This equation can be solved analytically ([Exercise 2.3.1](#)) but once again we prefer a graphical approach. We plot $\dot{N}$ versus N to see what the vector field looks like. Note that we plot only $N \geq 0$, since it makes no sense to think about a negative population ([Figure 2.3.3](#)).

Fixed points occur at $N^* = 0$ and $N^* = K$, as found by setting $\dot{N} = 0$ and solving for N . By looking at the flow in [Figure 2.3.3](#), we see that $N^* = 0$ is an unstable fixed point and $N^* = K$ is a stable fixed point. In biological terms,

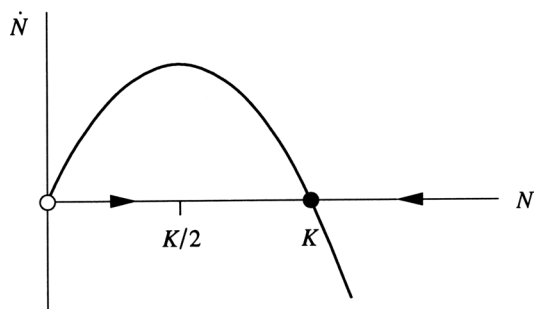


Figure 2.3.3

$N = 0$ is an unstable equilibrium: a small population will grow exponentially fast and run away from $N = 0$. On the other hand, if N is disturbed slightly from K , the disturbance will decay monotonically and $N(t) \rightarrow K$ as $t \rightarrow \infty$.

In fact, Figure 2.3.3 shows that if we start a phase point at *any* $N_0 > 0$, it will always flow toward $N = K$. Hence *the population always approaches the carrying capacity*.

The only exception is if $N_0 = 0$; then there's nobody around to start reproducing, and so $N = 0$ for all time. (The model does not allow for spontaneous generation!)

Figure 2.3.3 also allows us to deduce the qualitative shape of the solutions. For example, if $N_0 < K/2$, the phase point moves faster and faster until it crosses $N = K/2$, where the parabola in Figure 2.3.3 reaches its maximum. Then the phase point slows down and eventually creeps toward $N = K$. In biological terms, this means that the population initially grows in an accelerating fashion, and the graph of $N(t)$ is concave up. But after $N = K/2$, the derivative $\dot{N}$ begins to decrease, and so $N(t)$ is concave down as it asymptotes to the horizontal line $N = K$ (Figure 2.3.4). Thus the graph of $N(t)$ is S-shaped or *sigmoid* for $N_0 < K/2$.

Something qualitatively different occurs if the initial condition N_0 lies between $K/2$ and K ; now the solutions are decelerating from the start. Hence these solutions are concave down for all t . If the population initially exceeds the carrying capacity ($N_0 > K$), then $N(t)$ decreases toward $N = K$ and is concave up. Finally, if $N_0 = 0$ or $N_0 = K$, then the population stays constant.

Critique of the Logistic Model

Before leaving this example, we should make a few comments about the biological validity of the logistic equation. The algebraic form of the model is not to be taken literally. The model should really be regarded as a metaphor for populations that have a tendency to grow from zero population up to some carrying capacity K .

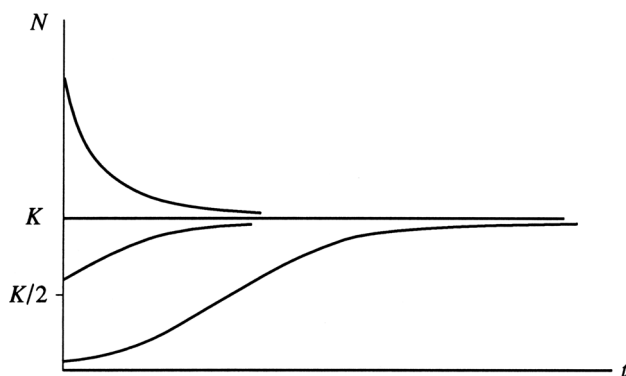


Figure 2.3.4

Originally a much stricter interpretation was proposed, and the model was argued to be a universal law of growth (Pearl 1927). The logistic equation was tested in laboratory experiments in which colonies of bacteria, yeast, or other simple organisms were grown in conditions of constant climate, food supply, and absence of predators. For a good review of this literature, see Krebs (1972, pp. 190–200). These experiments often yielded sigmoid growth curves, in some cases with an impressive match to the logistic predictions.

On the other hand, the agreement was much worse for fruit flies, flour beetles, and other organisms that have complex life cycles involving eggs, larvae, pupae, and adults. In these organisms, the predicted asymptotic approach to a steady carrying capacity was never observed—instead the populations exhibited large, persistent fluctuations after an initial period of logistic growth. See Krebs (1972) for a discussion of the possible causes of these fluctuations, including age structure and time-delayed effects of overcrowding in the population.

For further reading on population biology, see Pielou (1969) or May (1981). Edelstein–Keshet (1988) and Murray (2002, 2003) are excellent textbooks on mathematical biology in general.

2.4 Linear Stability Analysis

So far we have relied on graphical methods to determine the stability of fixed points. Frequently one would like to have a more quantitative measure of stability, such as the rate of decay to a stable fixed point. This sort of information may be obtained by *linearizing* about a fixed point, as we now explain.

Let x^* be a fixed point, and let

$$u(t) = x(t) - x^*$$

be a small perturbation away from x^* . To see whether the perturbation grows or decays, we derive a differential equation for u . Differentiation yields

$$\dot{u} = \frac{d}{dt}(x - x^*) = \dot{x},$$

since x^* is constant. Thus $\dot{u} = \dot{x} = f(x) = f(x^* + u)$. Now using Taylor's expansion we obtain

$$f(x^* + u) = f(x^*) + f'(x^*)u + O(u^2),$$

where $O(u^2)$ denotes quadratically small terms in u . Finally, observe that $f(x^*) = 0$ since x^* is a fixed point. Hence

$$\dot{u} = f'(x^*)u + O(u^2).$$

If $f'(x^*) \neq 0$, the $O(u^2)$ terms are negligible in comparison to the linear term, so we may write the approximation

$$\dot{u} \approx f'(x^*)u.$$

This linear differential equation for u is called the **linearization** about x^* . When pondering it, keep in mind that $f'(x^*)$ is just a glorified constant, since x^* is constant. The only thing that depends on time here is the perturbation $u(t)$. The linearization shows that $u(t)$ grows exponentially if $f'(x^*) > 0$ and decays exponentially if $f'(x^*) < 0$. If $f'(x^*) = 0$, the $O(u^2)$ terms are not negligible and a nonlinear analysis is needed to determine stability, as discussed in [Example 2.4.3](#) below.

The upshot is that the slope $f'(x^*)$ at the fixed point determines its stability. If you look back at the earlier examples, you'll see that the slope was always negative at a stable fixed point. The importance of the *sign* of $f'(x^*)$ was clear from our graphical approach; the new feature is that now we have a measure of *how* stable a fixed point is—that's determined by the *magnitude* of $f'(x^*)$. This magnitude plays the role of an exponential growth or decay rate. Its reciprocal $1/|f'(x^*)|$ is a **characteristic time scale**; it determines the time required for $x(t)$ to vary significantly in the neighborhood of x^* .

Example 2.4.1:

Using linear stability analysis, determine the stability of the fixed points for $\dot{x} = \sin x$.

Solution: The fixed points occur where $f(x) = \sin x = 0$. Thus $x^* = k\pi$, where k is an integer. Then

$$f'(x^*) = \cos k\pi = \begin{cases} 1, & k \text{ even} \\ -1, & k \text{ odd.} \end{cases}$$

Hence x^* is unstable if k is even and stable if k is odd. This agrees with the results shown in [Figure 2.1.1](#). ■

Example 2.4.2:

Classify the fixed points of the logistic equation, using linear stability analysis, and find the characteristic time scale in each case.

Solution: Here

$$f(N) = rN \left(1 - \frac{N}{K}\right)$$

with fixed points $N^* = 0$ and $N^* = K$. Then

$$f'(N) = r - \frac{2rN}{K}$$

and so $f'(0) = r$ and $f'(K) = -r$. Hence $N^* = 0$ is unstable and $N^* = K$ is stable, as found earlier by graphical arguments. In either case, the characteristic time scale is $1/|f'(N^*)| = 1/r$. ■

Example 2.4.3:

What can be said about the stability of a fixed point when $f'(x^*) = 0$?

Solution: Nothing can be said in general. The stability is best determined on a case-by-case basis, using graphical methods. Consider the following examples:

$$(a) \dot{x} = -x^3 \quad (b) \dot{x} = x^3 \quad (c) \dot{x} = x^2 \quad (d) \dot{x} = 0$$

Each of these systems has a fixed point $x^* = 0$ with $f'(x^*) = 0$. However the stability is different in each case. Figure 2.4.1 shows that (a) is stable and (b) is unstable. Case (c) is a hybrid case we'll call **half-stable**, since the fixed point is attracting from the left and repelling from the right. We therefore indicate this type of fixed point by a half-filled circle. Case (d) is a whole line of fixed points; perturbations neither grow nor decay.

These examples may seem artificial, but we will see that they arise naturally in the context of *bifurcations*—more about that later. ■

2.5 Existence and Uniqueness

Our treatment of vector fields has been very informal. In particular, we have taken a cavalier attitude toward questions of existence and uniqueness of solutions to the system $\dot{x} = f(x)$. That's in keeping with the “applied” spirit of this book. Nevertheless, we should be aware of what can go wrong in pathological cases.

Example 2.5.1:

Show that the solution to $\dot{x} = x^{1/3}$ starting from $x_0 = 0$ is *not* unique.

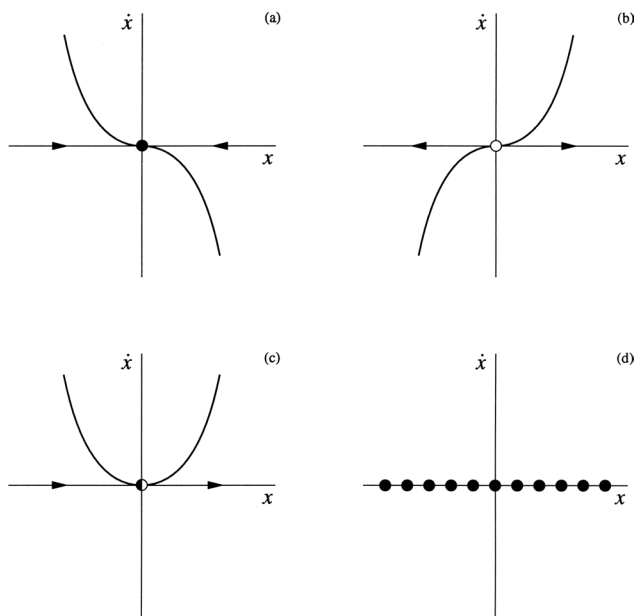


Figure 2.4.1

Solution: The point $x = 0$ is a fixed point, so one obvious solution is $x(t) = 0$ for all t . The surprising fact is that there is *another* solution. To find it we separate variables and integrate:

$$\int x^{-1/3} dx = \int dt$$

so $\frac{3}{2}x^{2/3} = t + C$. Imposing the initial condition $x(0) = 0$ yields $C = 0$. Hence $x(t) = (\frac{2}{3}t)^{3/2}$ is also a solution! ■

When uniqueness fails, our geometric approach collapses because the phase point doesn't know how to move; if a phase point were started at the origin, would it stay there or would it move according to $x(t) = (\frac{2}{3}t)^{3/2}$? (Or as my friends in elementary school used to say when discussing the problem of the irresistible force and the immovable object, perhaps the phase point would explode!)

Actually, the situation in [Example 2.5.1](#) is even worse than we've let on—there are *infinitely* many solutions starting from the same initial condition (Exercise 2.5.4).

What's the source of the non-uniqueness? A hint comes from looking at the vector field ([Figure 2.5.1](#)). We see that the fixed point $x^* = 0$ is *very* unstable—the slope $f'(0)$ is infinite.

Chastened by this example, we state a theorem that provides sufficient conditions for existence and uniqueness of solutions to $\dot{x} = f(x)$.

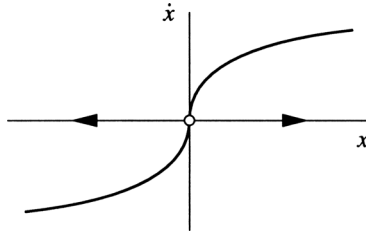


Figure 2.5.1

Existence and Uniqueness Theorem: Consider the initial value problem

$$\dot{x} = f(x), \quad x(0) = x_0.$$

Suppose that $f(x)$ and $f'(x)$ are continuous on an open interval R of the x -axis, and suppose that x_0 is a point in R . Then the initial value problem has a solution $x(t)$ on some time interval $(-\tau, \tau)$ about $t = 0$, and the solution is unique.

For proofs of the existence and uniqueness theorem, see Borrelli and Coleman (1987), Lin and Segel (1988), or virtually any text on ordinary differential equations.

This theorem says that *if* $f(x)$ is *smooth enough*, then solutions exist and are unique. Even so, there's no guarantee that solutions exist forever, as shown by the next example.

Example 2.5.2:

Discuss the existence and uniqueness of solutions to the initial value problem $\dot{x} = 1 + x^2$, $x(0) = x_0$. Do solutions exist for all time?

Solution: Here $f(x) = 1 + x^2$. This function is continuous and has a continuous derivative for all x . Hence the theorem tells us that solutions exist and are unique for any initial condition x_0 . But *the theorem does not say that the solutions exist for all time*; they are only guaranteed to exist in a (possibly very short) time interval around $t = 0$.

For example, consider the case where $x(0) = 0$. Then the problem can be solved analytically by separation of variables:

$$\int \frac{dx}{1+x^2} = \int dt,$$

which yields

$$\tan^{-1} x = t + C.$$

The initial condition $x(0) = 0$ implies $C = 0$. Hence $x(t) = \tan t$ is the solution. But notice that this solution exists only for $-\pi/2 < t < \pi/2$, because $x(t) \rightarrow \pm\infty$ as $t \rightarrow \pm\pi/2$. Outside of that time interval, there is no solution to the initial value problem for $x_0 = 0$. ■

The amazing thing about [Example 2.5.2](#) is that the system has solutions that reach infinity in *finite* time. This phenomenon is called **blow-up**. As the name suggests, it is of physical relevance in models of combustion and other runaway processes.

There are various ways to extend the existence and uniqueness theorem. One can allow f to depend on time t , or on several variables $x_1, \dots, x_n$. One of the most useful generalizations will be discussed later in [Section 6.2](#).

From now on, we will not worry about issues of existence and uniqueness—our vector fields will typically be smooth enough to avoid trouble. If we happen to come across a more dangerous example, we'll deal with it then.

2.6 Impossibility of Oscillations

Fixed points dominate the dynamics of first-order systems. In all our examples so far, all trajectories either approached a fixed point, or diverged to $\pm\infty$. In fact, those are the *only* things that can happen for a vector field on the real line. The reason is that trajectories are forced to increase or decrease monotonically, or remain constant ([Figure 2.6.1](#)). To put it more geometrically, the phase point never reverses direction.

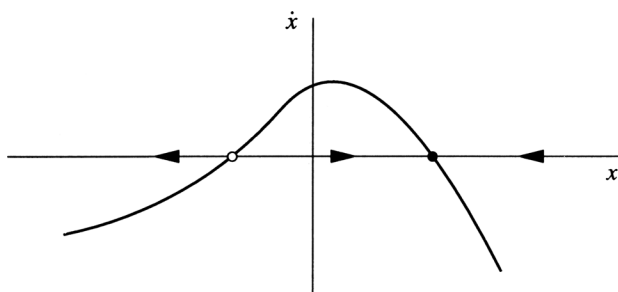


Figure 2.6.1

Thus, if a fixed point is regarded as an equilibrium solution, the approach to equilibrium is always *monotonic*; overshoot and damped oscillations can never occur in a first-order system. For the same reason, undamped oscillations are impossible. Hence *there are no periodic solutions to $\dot{x} = f(x)$* .

These general results are fundamentally topological in origin. They reflect the fact that $\dot{x} = f(x)$ corresponds to flow on a *line*. If you flow monotonically on a line, you'll never come back to your starting place—that's why periodic solutions are impossible. (Of course, if we were dealing with a *circle* rather than a line, we *could* eventually return to our starting place. Thus vector fields on the circle can exhibit periodic solutions, as we discuss in [Chapter 4](#).)

Mechanical Analog: Overdamped Systems

It may seem surprising that solutions to $\dot{x} = f(x)$ can't oscillate. But this result becomes obvious if we think in terms of a mechanical analog. We regard $\dot{x} = f(x)$ as a limiting case of Newton's law, in the limit where the "inertia term" $m\ddot{x}$ is negligible.

For example, suppose a mass m is attached to a nonlinear spring whose restoring force is $F(x)$, where x is the displacement from the origin. Furthermore, suppose that the mass is immersed in a vat of very viscous fluid, like honey or motor oil ([Figure 2.6.2](#)), so that it is subject to a damping force $b\dot{x}$. Then Newton's law is $m\ddot{x} + b\dot{x} = F(x)$.

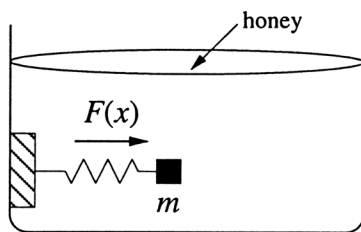


Figure 2.6.2

If the viscous damping is strong compared to the inertia term ($b\dot{x} \gg m\ddot{x}$), the system should behave like $b\dot{x} = F(x)$, or equivalently $\dot{x} = f(x)$, where $f(x) = b^{-1}F(x)$. In this **overdamped** limit, the behavior of the mechanical system is clear. The mass prefers to sit at a stable equilibrium, where $f(x) = 0$ and $f'(x) < 0$. If displaced a bit, the mass is slowly dragged back to equilibrium by the restoring force. No overshoot can occur, because the damping is enormous. And undamped oscillations are out of the question! These conclusions agree with those obtained earlier by geometric reasoning.

Actually, we should confess that this argument contains a slight swindle. The neglect of the inertia term $m\ddot{x}$ is valid, but only after a rapid initial transient during which the inertia and damping terms are of comparable size. An honest discussion of this point requires more machinery than we have available. We'll return to this matter in [Section 3.5](#).

2.7 Potentials

There's another way to visualize the dynamics of the first-order system $\dot{x} = f(x)$, based on the physical idea of potential energy. We picture a particle sliding down the walls of a potential well, where the **potential** $V(x)$ is defined by

$$f(x) = -\frac{dV}{dx}.$$

As before, you should imagine that the particle is heavily damped—its inertia is completely negligible compared to the damping force and the force due to the potential. For example, suppose that the particle has to slog through a thick layer of goo that covers the walls of the potential (Figure 2.7.1).

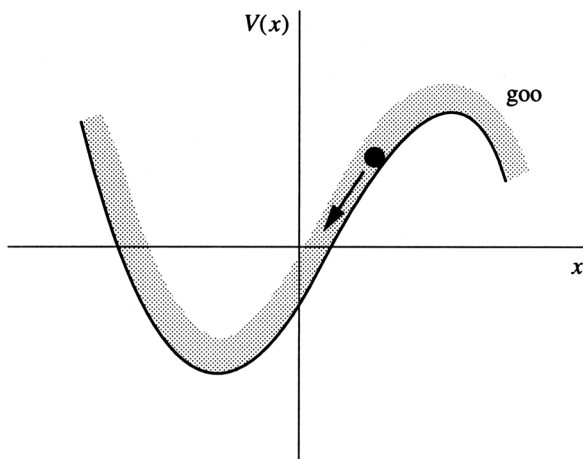


Figure 2.7.1

The negative sign in the definition of V follows the standard convention in physics; it implies that the particle always moves “downhill” as the motion proceeds. To see this, we think of x as a function of t , and then calculate the time-derivative of $V(x(t))$. Using the chain rule, we obtain

$$\frac{dV}{dt} = \frac{dV}{dx} \frac{dx}{dt}.$$

Now for a first-order system,

$$\frac{dx}{dt} = -\frac{dV}{dx},$$

since $\dot{x} = f(x) = -dV/dx$, by the definition of the potential. Hence,

$$\frac{dV}{dt} = -\left(\frac{dV}{dx}\right)^2 \leq 0.$$

Thus $V(t)$ *decreases along trajectories*, and so the particle always moves toward lower potential. Of course, if the particle happens to be at an **equilibrium** point where $dV/dx = 0$, then V remains constant. This is to be expected, since $dV/dx = 0$ implies $\dot{x} = 0$; equilibria occur at the fixed points

of the vector field. Note that local minima of $V(x)$ correspond to *stable* fixed points, as we'd expect intuitively, and local maxima correspond to *unstable* fixed points.

Example 2.7.1:

Graph the potential for the system $\dot{x} = -x$, and identify all the equilibrium points.

Solution: We need to find $V(x)$ such that $-dV/dx = -x$. The general solution is $V(x) = \frac{1}{2}x^2 + C$, where C is an arbitrary constant. (It always happens that the potential is only defined up to an additive constant. For convenience, we usually choose $C = 0$.) The graph of $V(x)$ is shown in Figure 2.7.2. The only equilibrium point occurs at $x = 0$, and it's stable. ■

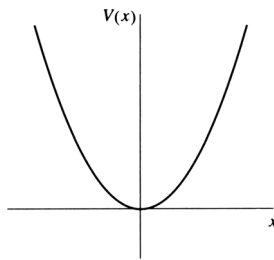


Figure 2.7.2

Example 2.7.2:

Graph the potential for the system $\dot{x} = x - x^3$, and identify all the equilibrium points.

Solution: Solving $-dV/dx = x - x^3$ yields $V = -\frac{1}{2}x^2 + \frac{1}{4}x^4 + C$. Once again we set $C = 0$. Figure 2.7.3 shows the graph of V . The local minima at $x = \pm 1$ correspond to stable equilibria, and the local maximum at $x = 0$ corresponds to an unstable equilibrium. The potential shown in Figure 2.7.3 is often called a **double-well potential**, and the system is said to be **bistable**, since it has two stable equilibria. ■

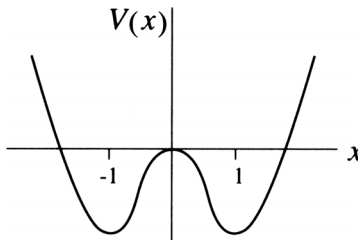


Figure 2.7.3

2.8 Solving Equations on the Computer

Throughout this chapter we have used graphical and analytical methods to analyze first-order systems. Every budding dynamicist should master a third tool: numerical methods. In the old days, numerical methods were impractical because they required enormous amounts of tedious hand-calculation. But all that has changed, thanks to the computer. Computers enable us to approximate the solutions to analytically intractable problems, and also to visualize those solutions. In this section we take our first look at dynamics on the computer, in the context of **numerical integration** of $\dot{x} = f(x)$.

Numerical integration is a vast subject. We will barely scratch the surface. See Chapter 17 of Press et al. (2007) for an excellent treatment.

Euler's Method

The problem can be posed this way: given the differential equation $\dot{x} = f(x)$, subject to the condition $x = x_0$ at $t = t_0$, find a systematic way to approximate the solution $x(t)$.

Suppose we use the vector field interpretation of $\dot{x} = f(x)$. That is, we think of a fluid flowing steadily on the x -axis, with velocity $f(x)$ at the location x . Imagine we're riding along with a phase point being carried downstream by the fluid. Initially we're at x_0 , and the local velocity is $f(x_0)$. If we flow for a short time Δt , we'll have moved a distance $f(x_0)\Delta t$, because distance = rate $\times$ time. Of course, that's not quite right, because our velocity was changing a little bit throughout the step. But over a sufficiently *small* step, the velocity will be nearly constant and our approximation should be reasonably good. Hence our new position $x(t_0 + \Delta t)$ is approximately $x_0 + f(x_0)\Delta t$. Let's call this approximation x_1 . Thus

$$x(t_0 + \Delta t) \approx x_1 = x_0 + f(x_0)\Delta t.$$

Now we iterate. Our approximation has taken us to a new location x_1 ; our new velocity is $f(x_1)$; we step forward to $x_2 = x_1 + f(x_1)\Delta t$; and so on. In general, the update rule is

$$x_{n+1} = x_n + f(x_n)\Delta t.$$

This is the simplest possible numerical integration scheme. It is known as **Euler's method**.

Euler's method can be visualized by plotting x versus t (Figure 2.8.1). The curve shows the exact solution $x(t)$, and the open dots show its values $x(t_n)$ at the discrete times $t_n = t_0 + n\Delta t$. The black dots show the approximate values given by the Euler method. As you can see, the approximation gets bad in a hurry unless Δt is extremely small. Hence Euler's method is not

recommended in practice, but it contains the conceptual essence of the more accurate methods to be discussed next.

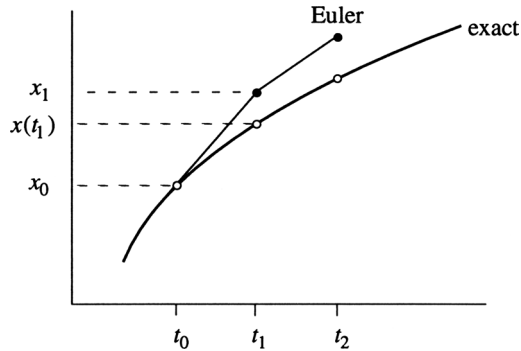


Figure 2.8.1

Refinements

One problem with the Euler method is that it estimates the derivative only at the left end of the time interval between t_n and t_{n+1} . A more sensible approach would be to use the *average* derivative across this interval. This is the idea behind the *improved Euler method*. We first take a trial step across the interval, using the Euler method. This produces a trial value $\tilde{x}_{n+1} = x_n + f(x_n)\Delta t$; the tilde above the x indicates that this is a tentative step, used only as a probe. Now that we've estimated the derivative on both ends of the interval, we average $f(x_n)$ and $f(\tilde{x}_{n+1})$, and use that to take the *real* step across the interval. Thus the improved Euler method is

$$\tilde{x}_{n+1} = x_n + f(x_n)\Delta t \quad (\text{the trial step})$$

$$x_{n+1} = x_n + \frac{1}{2} [f(x_n) + f(\tilde{x}_{n+1})] \Delta t. \quad (\text{the real step})$$

This method is more accurate than the Euler method, in the sense that it tends to make a smaller **error** $E = |x(t_n) - x_n|$ for a given **stepsize** Δt . In both cases, the error $E \rightarrow 0$ as $\Delta t \rightarrow 0$, but the error decreases *faster* for the improved Euler method. One can show that $E \propto \Delta t$ for the Euler method, but $E \propto (\Delta t)^2$ for the improved Euler method (Exercises 2.8.7 and 2.8.8). In the jargon of numerical analysis, the Euler method is first order, whereas the improved Euler method is second order.

Methods of third, fourth, and even higher orders have been concocted, but you should realize that higher-order methods are not necessarily superior. Higher-order methods require more calculations and function evaluations, so there's a computational cost associated with them. In practice, a good balance is achieved by the **fourth-order Runge-Kutta method**. To find x_{n+1}

in terms of x_n , this method first requires us to calculate the following four numbers (cunningly chosen, as you'll see in Exercise 2.8.9):

$$\begin{aligned}k_1 &= f(x_n)\Delta t \\k_2 &= f(x_n + \tfrac{1}{2}k_1)\Delta t \\k_3 &= f(x_n + \tfrac{1}{2}k_2)\Delta t \\k_4 &= f(x_n + k_3)\Delta t.\end{aligned}$$

Then x_{n+1} is given by

$$x_{n+1} = x_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

This method generally gives accurate results without requiring an excessively small stepsize Δt . Of course, some problems are nastier, and may require small steps in certain time intervals, while permitting very large steps elsewhere. In such cases, you may want to use a Runge–Kutta routine with an automatic stepsize control; see Press et al. (2007) for details.

Now that computers are so fast, you may wonder why we don't just pick a tiny Δt once and for all. The trouble is that excessively many computations will occur, and each one carries a penalty in the form of **round-off error**. Computers don't have infinite accuracy—they don't distinguish between numbers that differ by some small amount δ . For numbers of order 1, typically $\delta \approx 10^{-7}$ for single precision and $\delta \approx 10^{-16}$ for double precision. Round-off error occurs during every calculation, and will begin to accumulate in a serious way if Δt is too small. See Hubbard and West (1991) for a good discussion.

Practical Matters

You have several options if you want to solve differential equations on the computer. If you like to do things yourself, you can write your own numerical integration routines in your favorite programming language, and plot the results using whatever graphics programs are available. The information given above should be enough to get you started. For further guidance, consult Press et al. (2007).

A second option is to use existing packages. *MATLAB*, *Mathematica*, *Maple*, *Python* and *Sage* all have programs for solving ordinary differential equations and graphing their solutions.

The final option is for people who want to explore dynamics, not computing. Dynamical systems software is available for laptops, tablets, and smart phones. All you have to do is type in the equations and the parameters; the program solves the equations numerically and plots the results. Some recommended apps are *Slopes* (available for the iPhone and iPad as well as Android; this is a pleasant choice for beginners) and *XPPAUT* (by Bard Ermentrout, available on many platforms including iPhone and iPad; this is a much more powerful tool for researchers and serious users). There are also quite a few

solvers available online. They are of varying quality, but they are free, easy to use, and they run in the browser.

Example 2.8.1:

Solve the system $\dot{x} = x(1 - x)$ numerically.

Solution: This is a logistic equation (Section 2.3) with parameters $r = 1$, $K = 1$. Previously we gave a rough sketch of the solutions, based on geometric arguments; now we can draw a more quantitative picture.

As a first step, we plot the **slope field** for the system in the (t, x) plane (Figure 2.8.2). Here the equation $\dot{x} = x(1 - x)$ is being interpreted in a new way: for each point (t, x) , the equation gives the slope dx/dt of the solution passing through that point. The slopes are indicated by little line segments in Figure 2.8.2.

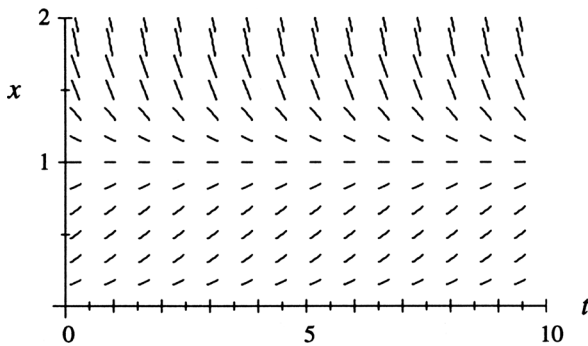


Figure 2.8.2

Finding a solution now becomes a problem of drawing a curve that is always tangent to the local slope. Figure 2.8.3 shows four solutions starting from various points in the (t, x) plane.

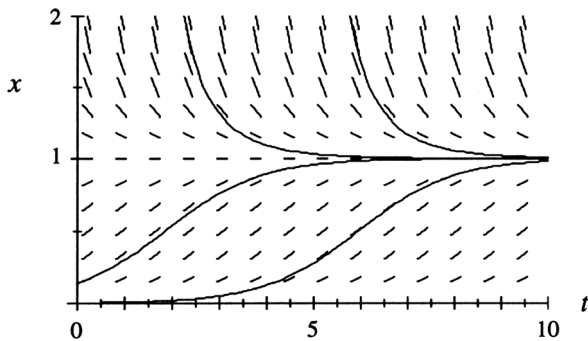


Figure 2.8.3

These numerical solutions were computed using the Runge–Kutta method with a stepsize $\Delta t = 0.1$. The solutions have the shape expected from [Section 2.3](#). ■

Computers are indispensable for studying dynamical systems. We will use them liberally throughout this book, and you should do likewise.

EXERCISES FOR CHAPTER 2

2.1 A Geometric Way of Thinking

In the next three exercises, interpret $\dot{x} = \sin x$ as a flow on the x -axis.

2.1.1 Find all the fixed points of the flow.

2.1.2 At which points x does the flow have greatest velocity to the right?

2.1.3 (Acceleration)

- a) Find the flow's acceleration $\ddot{x}$ as a function of x .
- b) Find the points where the flow has maximum positive acceleration.

2.1.4 (Exact solution of $\dot{x} = \sin x$) As shown in the text, $\dot{x} = \sin x$ has the solution $t = \ln |(\csc x_0 + \cot x_0)/(\csc x + \cot x)|$, where $x_0 = x(0)$ is the initial value of x .

- a) Given the specific initial condition $x_0 = \pi/4$, show that the solution above can be inverted to obtain

$$x(t) = 2 \tan^{-1} \left(\frac{e^t}{1 + \sqrt{2}} \right).$$

Conclude that $x(t) \rightarrow \pi$ as $t \rightarrow \infty$, claimed in [Section 2.1](#). (You need to be good with trigonometric identities to solve this problem.)

- b) Try to find the analytical solution for $x(t)$, given an *arbitrary* initial condition x_0 .

2.1.5 (A mechanical analog)

- a) Find a mechanical system that is approximately governed by $\dot{x} = \sin x$.
- b) Using your physical intuition, explain why it now becomes obvious that $x^* = 0$ is an unstable fixed point and $x^* = \pi$ is stable.

2.2 Fixed Points and Stability

Analyze the following equations graphically. In each case, sketch the vector field on the real line, find all the fixed points, classify their stability, and sketch the graph of $x(t)$ for different initial conditions. Then try for a few minutes to obtain the analytical solution for $x(t)$; if you get stuck, don't try for too long since in several cases it's impossible to solve the equation in closed form!

2.2.1 $\dot{x} = 4x^2 - 16$

2.2.2 $\dot{x} = 1 - x^{14}$

2.2.3 $\dot{x} = x - x^3$

2.2.4 $\dot{x} = e^{-x} \sin x$

2.2.5 $\dot{x} = 1 + \frac{1}{2} \cos x$

2.2.6 $\dot{x} = 1 - 2 \cos x$

2.2.7 $\dot{x} = e^x - \cos x$ (Hint: Sketch the graphs of e^x and $\cos x$ on the same axes, and look for intersections. You won't be able to find the fixed points explicitly, but you can still find the qualitative behavior.)

2.2.8 (Working backwards, from flows to equations) Given an equation $\dot{x} = f(x)$, we know how to sketch the corresponding flow on the real line. Here you are asked to solve the opposite problem: For the phase portrait shown in [Figure 1](#), find an equation that is consistent with it. (There are an infinite number of correct answers—and wrong ones too.)



Figure 1

2.2.9 (Backwards again, now from solutions to equations) Find an equation $\dot{x} = f(x)$ whose solutions $x(t)$ look like those shown in [Figure 2](#).

2.2.10 (Fixed points) For each of (a)–(e), find an equation $\dot{x} = f(x)$ with the stated properties, or if there are no examples, explain why not. (In all cases, assume that $f(x)$ is a smooth function.)

- Every real number is a fixed point.
- Every integer is a fixed point, and there are no others.
- There are precisely three fixed points, and all of them are stable.

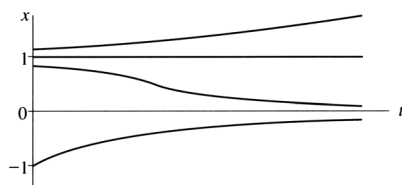


Figure 2

- d) There are no fixed points.
- e) There are precisely 100 fixed points.

2.2.11 (Analytical solution for charging capacitor) Obtain the analytical solution of the initial value problem

$$\dot{Q} = \frac{V_0}{R} - \frac{Q}{RC}$$

with $Q(0) = 0$, which arose in [Example 2.2.2](#).

2.2.12 (A nonlinear resistor) Suppose the resistor in [Example 2.2.2](#) is replaced by a nonlinear resistor. In other words, this resistor does not have a linear relation between voltage and current. Such nonlinearity arises in certain solid-state devices. Instead of $I_R = V/R$, suppose we have $I_R = g(V)$, where $g(V)$ has the shape shown in [Figure 3](#).

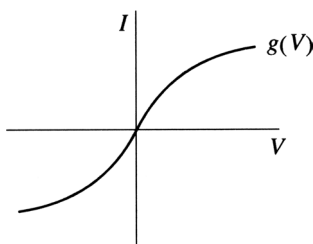


Figure 3

Redo [Example 2.2.2](#) in this case. Derive the circuit equations, find all the fixed points, and analyze their stability. What qualitative effects does the nonlinearity introduce (if any)?

2.2.13 (Terminal velocity) The velocity $v(t)$ of a skydiver falling to the ground is governed by

$$m\dot{v} = mg - kv^2$$

where m is the mass of the skydiver, g is the acceleration due to gravity, and $k > 0$ is a constant related to the amount of air resistance.

- a) Obtain the analytical solution for $v(t)$, assuming that $v(0) = 0$.
- b) Find the limit of $v(t)$ as $t \rightarrow \infty$. This limiting velocity is called the *terminal velocity*. (Beware of bad jokes about the word *terminal* and parachutes that fail to open.)
- c) Give a graphical analysis of this problem, and thereby re-derive a formula for the terminal velocity.
- d) An experimental study (Carlson et al. 1942) confirmed that the equation $m\dot{v} = mg - kv^2$ gives a good quantitative fit to data on human skydivers. Six men were dropped from altitudes varying from 10,600 feet to 31,400 feet to a terminal altitude of 2,100 feet, at which they opened their parachutes. The long free fall from 31,400 to 2,100 feet took 116 seconds. The average weight of the men and their equipment was 261.2 pounds. In these units, $g = 32.2 \text{ ft/sec}^2$. Compute the average velocity V_{avg} for the long free fall.
- e) Using the data given here, estimate the terminal velocity, and the value of the drag constant k . (Hints: First you need to find an exact formula for $s(t)$, the distance fallen, where $s(0) = 0$, $\dot{s} = v$, and $v(t)$ is known from part (a). You should get $s(t) = \frac{v^2}{g} \ln(\cosh \frac{gt}{V})$, where V is the terminal velocity. Then solve for V graphically or numerically, using $s = 29,300$, $t = 116$, and $g = 32.2$.)
 A slicker way to estimate V is to suppose $V \approx V_{\text{avg}}$ for the long free fall, as a rough first approximation. Then show that $gt/V \approx 15$. Since $gt/V \gg 1$, we may use the approximation $\ln(\cosh x) \approx x - \ln 2$ for $x \gg 1$. Derive this approximation and then use it to obtain an analytical estimate of V . Then k follows from part (b).

This analysis is from Davis (1962).

2.2.14 (Energy-balance model for Earth's temperature) Climate scientists use a hierarchy of mathematical models, ranging from very detailed to very simplified. At one end of the spectrum are comprehensive models containing billions of variables about the state of the atmosphere and oceans, also taking account of sea ice, terrestrial vegetation, ecosystems, biogeochemical cycles (such as the carbon cycle), and atmospheric chemistry. At the other extreme are simple conceptual models. These are a lot easier to understand, yet they can still provide valuable insights.

One simple model summarizes the climate in a single variable: the mean surface temperature of the Earth, averaged over the entire globe. The model ignores differences in the atmosphere's composition, as well as differences among continents and oceans, topography, and all other

local features. Its dynamics are governed by the following equation:

$$\begin{aligned} C \frac{dT}{dt} &= E_{\text{in}} - E_{\text{out}} \\ &= (1 - \alpha)Q - \epsilon \sigma T^4. \end{aligned}$$

Here C is the heat capacity of the Earth, and T is the mean surface temperature (measured in degrees Kelvin) at time t . This is called an *energy-balance model* because it assumes the mean temperature changes in response to the amount of energy reaching the Earth from the Sun (E_{in} , proportional to the solar irradiance parameter Q) minus the energy emitted back out into the stratosphere (E_{out} , assumed to follow the Stefan-Boltzmann law). The parameter $0 < \alpha < 1$ is called the *albedo*; it's the fraction of incoming solar radiation that the Earth reflects. Its numerical value is different for ice, water, and land, but those details are averaged out in this simplified model. Likewise, $0 < \epsilon < 1$ accounts in an averaged way for the fact that some of the outgoing radiative energy does not make it to outer space; instead, a fraction of it gets absorbed by greenhouse effects in the atmosphere.

- Find the model's prediction for Earth's mean surface temperature T^* in steady state, expressed in terms of the other parameters.
- Show graphically that T^* is a *stable* fixed point.
- Sketch T^* versus Q .
- Estimate the numerical value of T^* in degrees Kelvin. (You may assume $Q = 342 \text{ Watts}/(\text{meter})^2$, $\alpha \approx 0.3$, $\sigma = 5.67 \times 10^{-8} \text{ Watts}/(\text{meter})^2 \text{K}^4$, and $\epsilon \approx 0.62$.)

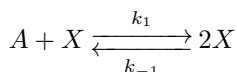
This analysis is from Kaper and Engler (2013).

2.3 Population Growth

2.3.1 (Exact solution of logistic equation) There are two ways to solve the logistic equation $\dot{N} = rN(1 - N/K)$ analytically for an arbitrary initial condition N_0 .

- Separate variables and integrate, using partial fractions.
- Make the change of variables $x = 1/N$. Then derive and solve the resulting differential equation for x .

2.3.2 (Autocatalysis) Consider the model chemical reaction



in which one molecule of X combines with one molecule of A to form two molecules of X . This means that the chemical X stimulates its own production, a process called *autocatalysis*. This positive feedback process leads to a chain reaction, which eventually is limited by a “back reaction” in which $2X$ returns to $A + X$.

According to the *law of mass action* of chemical kinetics, the rate of an elementary reaction is proportional to the product of the concentrations of the reactants. We denote the concentrations by lowercase letters $x = [X]$ and $a = [A]$. Assume that there's an enormous surplus of chemical A , so that its concentration a can be regarded as constant. Then the equation for the kinetics of x is

$$\dot{x} = k_1 ax - k_{-1} x^2$$

where k_1 and k_{-1} are positive parameters called *rate constants*.

- a) Find all the fixed points of this equation and classify their stability.
- b) Sketch the graph of $x(t)$ for various initial values x_0 .

2.3.3 (Tumor growth) The growth of cancerous tumors can be modeled by the Gompertz law $\dot{N} = -aN \ln(bN)$, where $N(t)$ is proportional to the number of cells in the tumor, and $a, b > 0$ are parameters.

- a) Interpret a and b biologically.
- b) Sketch the vector field and graph $N(t)$ for various initial values.

The predictions of this simple model agree surprisingly well with data on tumor growth, as long as N is not too small; see Aroesty et al. (1973) and Newton (1980) for examples.

2.3.4 (The Allee effect) For certain species of organisms, the effective growth rate $\dot{N}/N$ is highest at intermediate N . This is called the Allee effect (Edelstein–Keshet 1988). For example, imagine that it is too hard to find mates when N is very small, and there is too much competition for food and other resources when N is large.

- a) Show that $\dot{N}/N = r - a(N - b)^2$ provides an example of the Allee effect, if r, a , and b satisfy certain constraints, to be determined.
- b) Find all the fixed points of the system and classify their stability.
- c) Sketch the solutions $N(t)$ for different initial conditions.
- d) Compare the solutions $N(t)$ to those found for the logistic equation. What are the qualitative differences, if any?

2.3.5 (Dominance of the fittest) Suppose X and Y are two species that reproduce exponentially fast: $\dot{X} = aX$ and $\dot{Y} = bY$, respectively, with initial conditions $X_0, Y_0 > 0$ and growth rates $a > b > 0$. Here X is “fitter” than Y in the sense that it reproduces faster, as reflected by the inequality $a > b$. So we’d expect X to keep increasing its share of the total population $X + Y$ as $t \rightarrow \infty$. The goal of this exercise is to demonstrate this intuitive result, first analytically and then geometrically.

- a) Let $x(t) = X(t)/[X(t) + Y(t)]$ denote X ’s share of the total population. By solving for $X(t)$ and $Y(t)$, show that $x(t)$ increases monotonically and approaches 1 as $t \rightarrow \infty$.
- b) Alternatively, we can arrive at the same conclusions by deriving a differential equation for $x(t)$. To do so, take the time derivative of $x(t) = X(t)/[X(t) + Y(t)]$ using the quotient and chain rules. Then substitute for $\dot{X}$ and $\dot{Y}$ and thereby show that $x(t)$ obeys the logistic equation $\dot{x} = (a - b)x(1 - x)$. Explain why this implies that $x(t)$ increases monotonically and approaches 1 as $t \rightarrow \infty$.

2.3.6 (Language death) Thousands of the world’s languages are vanishing at an alarming rate, with 90 percent of them being expected to disappear by the end of this century. Abrams and Strogatz (2003) proposed the following model of language competition, and compared it to historical data on the decline of Welsh, Scottish Gaelic, Quechua (the most common surviving indigenous language in the Americas), and other endangered languages.

Let X and Y denote two languages competing for speakers in a given society. The proportion of the population speaking X evolves according to

$$\dot{x} = (1 - x)P_{YX} - xP_{XY}$$

where $0 \leq x \leq 1$ is the current fraction of the population speaking X , $1 - x$ is the complementary fraction speaking Y , and P_{YX} is the rate at which individuals switch from Y to X . This deliberately idealized model assumes that the population is well mixed (meaning that it lacks all spatial and social structure) and that all speakers are monolingual.

Next, the model posits that the attractiveness of a language increases with both its number of speakers and its perceived status, as quantified by a parameter $0 \leq s \leq 1$ that reflects the social or economic opportunities afforded to its speakers. Specifically, assume that $P_{YX} = sx^a$ and, by symmetry, $P_{XY} = (1 - s)(1 - x)^a$, where the exponent $a > 1$ is an adjustable parameter. Then the model becomes

$$\dot{x} = s(1 - x)x^a - (1 - s)x(1 - x)^a.$$

- a) Show that this equation for $\dot{x}$ has three fixed points.

- b) Show that for all $a > 1$, the fixed points at $x = 0$ and $x = 1$ are both stable.
- c) Show that the third fixed point, $0 < x^* < 1$, is unstable.

This model therefore predicts that two languages cannot coexist stably—one will eventually drive the other to extinction. For a review of generalizations of the model that allow for bilingualism, social structure, etc., see Castellano et al. (2009).

2.4 Linear Stability Analysis

Use linear stability analysis to classify the fixed points of the following systems. If linear stability analysis fails because $f'(x^*) = 0$, use a graphical argument to decide the stability.

2.4.1 $\dot{x} = x(1 - x)$

2.4.2 $\dot{x} = x(1 - x)(2 - x)$

2.4.3 $\dot{x} = \tan x$

2.4.4 $\dot{x} = x^2(6 - x)$

2.4.5 $\dot{x} = 1 - e^{-x^2}$

2.4.6 $\dot{x} = \ln x$

2.4.7 $\dot{x} = ax - x^3$, where a can be positive, negative, or zero. Discuss all three cases.

2.4.8 Using linear stability analysis, classify the fixed points of the Gompertz model of tumor growth $\dot{N} = -aN \ln(bN)$. (As in Exercise 2.3.3, $N(t)$ is proportional to the number of cells in the tumor and $a, b > 0$ are parameters.)

2.4.9 (Critical slowing down) In statistical mechanics, the phenomenon of “critical slowing down” is a signature of a second-order phase transition. At the transition, the system relaxes to equilibrium much more slowly than usual. Here’s a mathematical version of the effect:

- a) Obtain the analytical solution to $\dot{x} = -x^3$ for an arbitrary initial condition. Show that $x(t) \rightarrow 0$ as $t \rightarrow \infty$, but that the decay is not exponential. (You should find that the decay is a much slower algebraic function of t .)

- b) To get some intuition about the slowness of the decay, make a numerically accurate plot of the solution for the initial condition $x_0 = 10$, for $0 \leq t \leq 10$. Then, on the same graph, plot the solution to $\dot{x} = -x$ for the same initial condition.

2.5 Existence and Uniqueness

2.5.1 (Reaching a fixed point in a finite time) A particle travels on the half-line $x \geq 0$ with a velocity given by $\dot{x} = -x^c$, where c is real and constant.

- a) Find all values of c such that the origin $x = 0$ is a stable fixed point.
- b) Now assume that c is chosen such that $x = 0$ is stable. Can the particle ever reach the origin in a *finite* time? Specifically, how long does it take for the particle to travel from $x = 1$ to $x = 0$, as a function of c ?

2.5.2 (“Blow-up”: Reaching infinity in a finite time) Show that the solution to $\dot{x} = 1 + x^{10}$ escapes to $+\infty$ in a finite time, starting from any initial condition. (Hint: Don’t try to find an exact solution; instead, compare the solutions to those of $\dot{x} = 1 + x^2$.)

2.5.3 Consider the equation $\dot{x} = rx + x^3$, where $r > 0$ is fixed. Show that $x(t) \rightarrow \pm\infty$ in finite time, starting from any initial condition $x_0 \neq 0$.

2.5.4 (Infinitely many solutions with the same initial condition) Show that the initial value problem $\dot{x} = x^{1/3}$, $x(0) = 0$, has an infinite number of solutions. (Hint: Construct a solution that stays at $x = 0$ until some arbitrary time t_0 , after which it takes off.)

2.5.5 (A general example of non-uniqueness) Consider the initial value problem $\dot{x} = |x|^{p/q}$, $x(0) = 0$, where p and q are positive integers with no common factors.

- a) Show that there are an infinite number of solutions for $x(t)$ if $p < q$.
- b) Show that there is a unique solution if $p > q$.

2.5.6 (The leaky bucket) The following example (Hubbard and West 1991, p. 159) shows that in some physical situations, non-uniqueness is natural and obvious, not pathological.

Consider a water bucket with a hole in the bottom. If you see an empty bucket with a puddle beneath it, can you figure out when the bucket

was full? No, of course not! It could have finished emptying a minute ago, ten minutes ago, or whatever. The solution to the corresponding differential equation must be non-unique when integrated backwards in time.

Here's a crude model of the situation. Let $h(t)$ = height of the water remaining in the bucket at time t ; a = area of the hole; A = cross-sectional area of the bucket (assumed constant); $v(t)$ = velocity of the water passing through the hole.

- a) Show that $av(t) = A\dot{h}(t)$. What physical law are you invoking?
- b) To derive an additional equation, use conservation of energy. First, find the change in potential energy in the system, assuming that the height of the water in the bucket decreases by an amount Δh and that the water has density ρ . Then find the kinetic energy transported out of the bucket by the escaping water. Finally, assuming all the potential energy is converted into kinetic energy, derive the equation $v^2 = 2gh$.
- c) Combining (a) and (b), show $\dot{h} = -C\sqrt{h}$, where $C = \sqrt{2g}(a/A)$.
- d) Given $h(0) = 0$ (bucket empty at $t = 0$), show that the solution for $h(t)$ is non-unique *in backwards time*, i.e., for $t < 0$.

2.6 Impossibility of Oscillations

2.6.1 Explain this paradox: a simple harmonic oscillator $m\ddot{x} = -kx$ is a system that oscillates in one dimension (along the x -axis). But the text says one-dimensional systems can't oscillate.

2.6.2 (No periodic solutions to $\dot{x} = f(x)$) Here's an analytic proof that periodic solutions are impossible for a vector field on a line. Suppose on the contrary that $x(t)$ is a nontrivial periodic solution, i.e., $x(t) = x(t + T)$ for some $T > 0$, and $x(t) \neq x(t + s)$ for all $0 < s < T$. Derive a contradiction by considering $\int_t^{t+T} f(x) \frac{dx}{dt} dt$.

2.7 Potentials

For each of the following vector fields, plot the potential function $V(x)$ and identify all the equilibrium points and their stability.

2.7.1 $\dot{x} = x(1 - x)$

2.7.2 $\dot{x} = 3$

2.7.3 $\dot{x} = \sin x$

2.7.4 $\dot{x} = 2 + \sin x$

2.7.5 $\dot{x} = -\sinh x$

2.7.6 $\dot{x} = r + x - x^3$, for various values of r .

2.7.7 (Another proof that solutions to $\dot{x} = f(x)$ can't oscillate) Let $\dot{x} = f(x)$ be a vector field on the line. Use the existence of a potential function $V(x)$ to show that solutions $x(t)$ cannot oscillate.

2.8 Solving Equations on the Computer

2.8.1 (Slope field) The slope is constant along horizontal lines in [Figure 2.8.2](#). Why should we have expected this?

2.8.2 Sketch the slope field for the following differential equations. Then “integrate” the equation manually by drawing trajectories that are everywhere parallel to the local slope.

a) $\dot{x} = x$

b) $\dot{x} = 1 - x^2$

c) $\dot{x} = 1 - 4x(1 - x)$

d) $\dot{x} = \sin x$

2.8.3 (Calibrating the Euler method) The goal of this problem is to test the Euler method on the initial value problem $\dot{x} = -x$, $x(0) = 1$.

- Solve the problem analytically. What is the exact value of $x(1)$?
- Using the Euler method with stepsize $\Delta t = 1$, estimate $x(1)$ numerically, and call the result $\hat{x}(1)$. Then repeat, using $\Delta t = 10^{-n}$, for $n = 1, 2, 3, 4$.
- Plot the error $E = |\hat{x}(1) - x(1)|$ as a function of Δt . Then plot $\ln(E)$ vs. $\ln(\Delta t)$. Explain the results.

2.8.4 Redo Exercise 2.8.3, using the improved Euler method.

2.8.5 Redo Exercise 2.8.3, using the Runge–Kutta method.

2.8.6 (Analytically intractable problem) Consider the initial value problem $\dot{x} = x + e^{-x}$, $x(0) = 0$. In contrast to Exercise 2.8.3, this problem can't be solved analytically.

- a) Sketch the solution $x(t)$ for $t \geq 0$.
- b) Using some analytical arguments, obtain rigorous bounds on the value of x at $t = 1$. In other words, prove that $a < x(1) < b$, for a, b to be determined. By being clever, try to make a and b as close together as possible. (Hint: Bound the given vector field by approximate vector fields that can be integrated analytically.)
- c) Now for the numerical part: Using the Euler method, compute x at $t = 1$, correct to three decimal places. How small does the step-size need to be to obtain the desired accuracy? (Give the order of magnitude, not the exact number.)
- d) Repeat part (c), now using the Runge–Kutta method. Compare the results for stepsizes $\Delta t = 1$, $\Delta t = 0.1$, and $\Delta t = 0.01$.

2.8.7 (Error estimate for Euler method) In this question you'll use Taylor series expansions to estimate the error in taking one step by the Euler method. The exact solution and the Euler approximation both start at $x = x_0$ when $t = t_0$. We want to compare the exact value $x(t_1) \equiv x(t_0 + \Delta t)$ with the Euler approximation $x_1 = x_0 + f(x_0)\Delta t$.

- a) Expand $x(t_1) = x(t_0 + \Delta t)$ as a Taylor series in Δt , through terms of $O(\Delta t^2)$. Express your answer solely in terms of x_0 , Δt , and f and its derivatives at x_0 .
- b) Show that the local error $|x(t_1) - x_1| \sim C\Delta t^2$ and give an explicit expression for the constant C . (Generally one is more interested in the global error incurred after integrating over a time interval of fixed length $T = n\Delta t$. Since each step produces an $O(\Delta t^2)$ error, and we take $n = T/\Delta t = O(\Delta t^{-1})$ steps, the global error $|x(t_n) - x_n|$ is $O(\Delta t)$, as claimed in the text.)

2.8.8 (Error estimate for the improved Euler method) Use the Taylor series arguments of Exercise 2.8.7 to show that the local error for the improved Euler method is $O(\Delta t^3)$.

2.8.9 (Error estimate for Runge–Kutta) Show that the Runge–Kutta method produces a local error of size $O(\Delta t^5)$. (Warning: This calculation involves massive amounts of algebra, but if you do it correctly, you'll be rewarded by seeing many wonderful cancellations. Teach yourself *Mathematica*, *Maple*, *Sage*, *Maxima*, or some other computer algebra system, and do the problem on the computer.)



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Chapter 3

Bifurcations

3.0 Introduction

As we've seen in [Chapter 2](#), the dynamics of vector fields on the line is very limited: all solutions either settle down to equilibrium or head out to $\pm\infty$. Given the triviality of the dynamics, what's interesting about one-dimensional systems? Answer: *Dependence on parameters*. The qualitative structure of the flow can change as parameters are varied. In particular, fixed points can be created or destroyed, or their stability can change. These qualitative changes in the dynamics are called **bifurcations**, and the parameter values at which they occur are called **bifurcation points**.

Bifurcations are important scientifically—they provide models of transitions and instabilities as some *control parameter* is varied. For example, consider the buckling of a beam. If a small weight is placed on top of the beam in [Figure 3.0.1](#), the beam can support the load and remain vertical. But if the load is too heavy, the vertical position becomes unstable, and the beam may buckle.

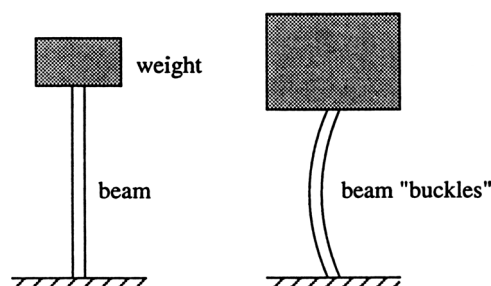


Figure 3.0.1

Here the weight plays the role of the control parameter, and the deflection of the beam from vertical plays the role of the dynamical variable x .

One of the main goals of this book is to help you develop a solid and practical understanding of bifurcations. This chapter introduces the simplest examples: bifurcations of fixed points for flows on the line. We'll use these bifurcations to model such dramatic phenomena as the onset of coherent

radiation in a laser and the outbreak of an insect population. (In later chapters, when we step up to two- and three-dimensional phase spaces, we'll explore additional types of bifurcations and their scientific applications.)

We begin with the most fundamental bifurcation of all.

3.1 Saddle-Node Bifurcation

The saddle-node bifurcation is the basic mechanism by which fixed points are *created and destroyed*. As a parameter is varied, two fixed points move toward each other, collide, and mutually annihilate.

The prototypical example of a saddle-node bifurcation is given by the first-order system

$$\dot{x} = r + x^2 \quad (1)$$

where r is a parameter, which may be positive, negative, or zero. When r is negative, there are two fixed points, one stable and one unstable (Figure 3.1.1a).

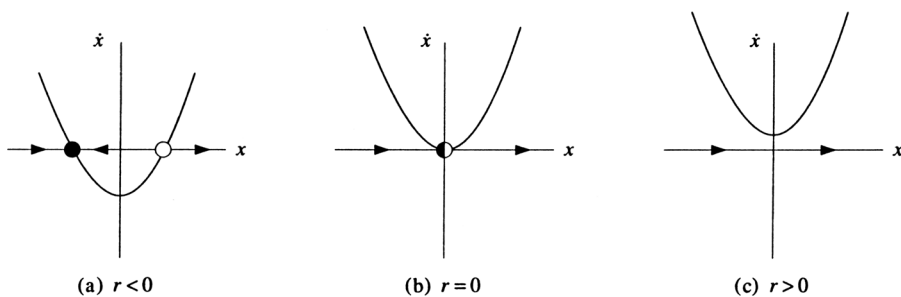


Figure 3.1.1

As r approaches 0 from below, the parabola moves up and the two fixed points move toward each other. When $r = 0$, the fixed points coalesce into a half-stable fixed point at $x^* = 0$ (Figure 3.1.1b). This type of fixed point is extremely delicate—it vanishes as soon as $r > 0$, and now there are no fixed points at all (Figure 3.1.1c).

In this example, we say that a *bifurcation* occurred at $r = 0$, since the vector fields for $r < 0$ and $r > 0$ are qualitatively different.

Graphical Conventions

There are several other ways to depict a saddle-node bifurcation. We can show a stack of vector fields for discrete values of r (Figure 3.1.2).

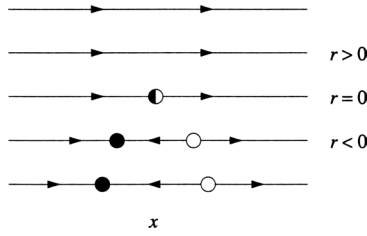


Figure 3.1.2

This representation emphasizes the dependence of the fixed points on r . In the limit of a *continuous* stack of vector fields, we have a picture like [Figure 3.1.3](#).

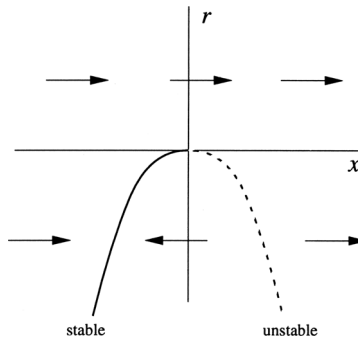


Figure 3.1.3

The curve shown is $r = -x^2$, i.e., $\dot{x} = 0$, which gives the fixed points for different r . To distinguish between stable and unstable fixed points, we use a solid line for stable points and a broken line for unstable ones.

However, the most common way to depict the bifurcation is to invert the axes of [Figure 3.1.3](#). The rationale is that r plays the role of an independent variable, and so should be plotted horizontally ([Figure 3.1.4](#)).

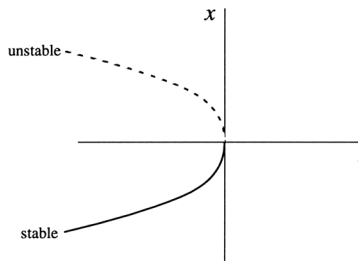


Figure 3.1.4

The drawback is that now the x -axis has to be plotted vertically, which looks strange at first. Arrows are sometimes included in the picture, but not always. This picture is called the **bifurcation diagram** for the saddle-node bifurcation.

Terminology

Bifurcation theory is rife with conflicting terminology. Some people refer to a saddle-node bifurcation as a *fold bifurcation* (because the curve in Figure 3.1.4 has a fold in it) or a *turning-point bifurcation* (because the point $(x, r) = (0, 0)$ is a “turning point.”) Admittedly, the term *saddle-node* doesn’t make much sense for vector fields on the line. The name derives from a completely analogous bifurcation seen in a higher-dimensional context, such as vector fields on the plane, where fixed points known as saddles and nodes can collide and annihilate (see Section 8.1).

The prize for most inventive terminology must go to Abraham and Shaw (1988), who write of a **blue sky bifurcation**. This term comes from viewing a saddle-node bifurcation in the other direction: a pair of fixed points appears “out of the clear blue sky” as a parameter is varied. For example, the vector field

$$\dot{x} = r - x^2 \quad (2)$$

has no fixed points for $r < 0$, but then one materializes when $r = 0$ and splits into two when $r > 0$ (Figure 3.1.5). Incidentally, this example also explains why we use the word “bifurcation”: it means “splitting into two branches.”

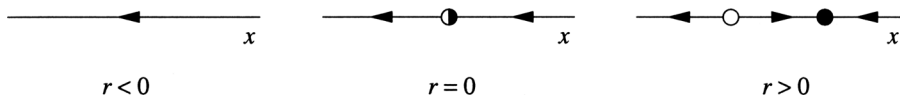


Figure 3.1.5

Example 3.1.1:

Give a linear stability analysis of the fixed points in Figure 3.1.5.

Solution: The fixed points for $\dot{x} = f(x) = r - x^2$ are given by $x^* = \pm\sqrt{r}$. There are two fixed points for $r > 0$, and none for $r < 0$. To determine linear stability, we compute $f'(x^*) = -2x^*$. Thus $x^* = +\sqrt{r}$ is stable, since $f'(x^*) < 0$. Similarly $x^* = -\sqrt{r}$ is unstable. At the bifurcation point $r = 0$, we find $f'(x^*) = 0$; the linearization vanishes when the fixed points coalesce. ■

Example 3.1.2:

Show that the first-order system $\dot{x} = r - x - e^{-x}$ undergoes a saddle-node bifurcation as r is varied, and find the value of r at the bifurcation point.

Solution: The fixed points satisfy $f(x) = r - x - e^{-x} = 0$. But now we run into a difficulty: in contrast to Example 3.1.1, we can’t find the fixed points

explicitly as a function of r . Instead we adopt a geometric approach. One method would be to graph the function $f(x) = r - x - e^{-x}$ for different values of r , look for its roots x^* , and then sketch the vector field on the x -axis. This method is fine, but there's an easier way. The point is that the two functions $r - x$ and e^{-x} have much more familiar graphs than their difference $r - x - e^{-x}$. So we plot $r - x$ and e^{-x} on the same picture (Figure 3.1.6a).

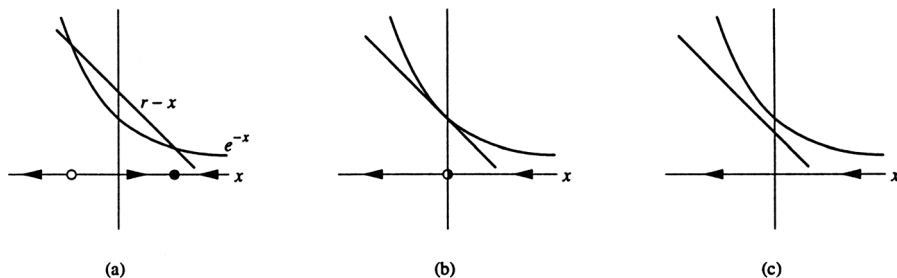


Figure 3.1.6

At points where the line $r - x$ intersects the curve e^{-x} , we have $r - x = e^{-x}$ and so $f(x) = 0$. Thus, *intersections of the line and the curve correspond to fixed points for the system*. This picture also allows us to read off the direction of flow on the x -axis: the flow is to the right where the line lies above the curve, since $r - x > e^{-x}$ and therefore $\dot{x} > 0$. Hence, the fixed point on the right is stable, and the one on the left is unstable.

Now imagine we start decreasing the parameter r . The line $r - x$ slides down and the fixed points approach each other. At some critical value $r = r_c$ the line becomes *tangent* to the curve and the fixed points coalesce in a saddle-node bifurcation (Figure 3.1.6b). For r below this critical value, the line lies below the curve and there are no fixed points (Figure 3.1.6c).

To find the bifurcation point r_c , we impose the condition that the graphs of $r - x$ and e^{-x} intersect *tangentially*. Thus we demand equality of the functions and their derivatives:

$$e^{-x} = r - x$$

and

$$\frac{d}{dx}(e^{-x}) = \frac{d}{dx}(r - x).$$

The second equation implies $-e^{-x} = -1$, so $x = 0$. Then the first equation yields $r = 1$. Hence the bifurcation point is $r_c = 1$, and the bifurcation occurs at $x = 0$. ■

Normal Forms

In a certain sense, the examples $\dot{x} = r - x^2$ or $\dot{x} = r + x^2$ are representative of *all* saddle-node bifurcations; that's why we called them "prototypical." The

idea is that, close to a saddle-node bifurcation, the dynamics typically look like $\dot{x} = r - x^2$ or $\dot{x} = r + x^2$.

For instance, consider [Example 3.1.2](#) near the bifurcation at $x = 0$ and $r = 1$. Using the Taylor expansion for e^{-x} about $x = 0$, we find

$$\begin{aligned}\dot{x} &= r - x - e^{-x} \\ &= r - x - \left[1 - x + \frac{x^2}{2!} + \cdots \right] \\ &= (r - 1) - \frac{1}{2}x^2 + \cdots\end{aligned}$$

to leading order in x . This has the same algebraic form as $\dot{x} = r - x^2$, and can be made to agree exactly by appropriate rescalings of x and r .

It's easy to understand why saddle-node bifurcations typically have this algebraic form. We just ask ourselves: how can two fixed points of $\dot{x} = f(x)$ collide and disappear as a parameter r is varied? Graphically, fixed points occur where the graph of $f(x)$ intersects the x -axis. For a saddle-node bifurcation to be possible, we need two nearby roots of $f(x)$; this means $f(x)$ must look locally “bowl-shaped” or parabolic ([Figure 3.1.7](#)).

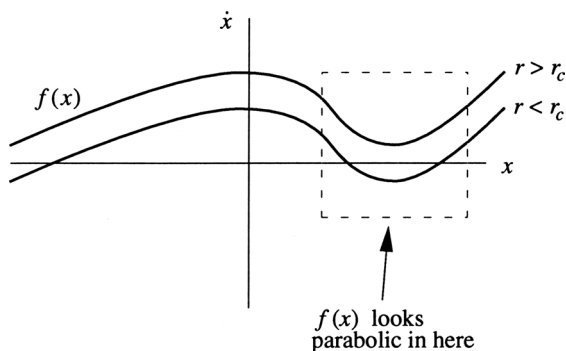


Figure 3.1.7

Now we use a microscope to zoom in on the behavior near the bifurcation. As r varies, we see a parabola intersecting the x -axis, then becoming tangent to it, and then failing to intersect. This is exactly the scenario in the prototypical [Figure 3.1.1](#).

Here's a more algebraic version of the same argument. We regard f as a function of both x and r , and examine the behavior of $\dot{x} = f(x, r)$ near the

bifurcation at $x = x^*$ and $r = r_c$. Taylor's expansion yields

$$\begin{aligned}\dot{x} &= f(x, r) \\ &= f(x^*, r_c) + (x - x^*) \left. \frac{\partial f}{\partial x} \right|_{(x^*, r_c)} + (r - r_c) \left. \frac{\partial f}{\partial r} \right|_{(x^*, r_c)} \\ &\quad + \frac{1}{2} (x - x^*)^2 \left. \frac{\partial^2 f}{\partial x^2} \right|_{(x^*, r_c)} + \dots\end{aligned}$$

where we have neglected quadratic terms in $(r - r_c)$ and cubic terms in $(x - x^*)$. Two of the terms in this equation vanish: $f(x^*, r_c) = 0$ since x^* is a fixed point, and $\partial f / \partial x|_{(x^*, r_c)} = 0$ by the tangency condition of a saddle-node bifurcation. Thus

$$\dot{x} = a(r - r_c) + b(x - x^*)^2 + \dots \quad (3)$$

where $a = \partial f / \partial r|_{(x^*, r_c)}$ and $b = \frac{1}{2} \partial^2 f / \partial x^2|_{(x^*, r_c)}$. Equation (3) agrees with the form of our prototypical examples. (We are assuming that $a, b \neq 0$, which is the typical case; for instance, it would be a very special situation if the second derivative $\partial^2 f / \partial x^2$ also happened to vanish at the fixed point.)

What we have been calling prototypical examples are more conventionally known as **normal forms** for the saddle-node bifurcation. There is much, much more to normal forms than we have indicated here. We will be seeing their importance throughout this book. For a more detailed and precise discussion, see Guckenheimer and Holmes (1983) or Wiggins (1990).

3.2 Transcritical Bifurcation

There are certain scientific situations where a fixed point must exist for all values of a parameter and can never be destroyed. For example, in the logistic equation and other simple models for the growth of a single species, there is a fixed point at zero population, regardless of the value of the growth rate. However, such a fixed point may *change its stability* as the parameter is varied. The transcritical bifurcation is the standard mechanism for such changes in stability.

The normal form for a transcritical bifurcation is

$$\dot{x} = rx - x^2. \quad (1)$$

This looks like the logistic equation of [Section 2.3](#), but now we allow x and r to be either positive or negative.

[Figure 3.2.1](#) shows the vector field as r varies. Note that there is a fixed point at $x^* = 0$ for *all* values of r .

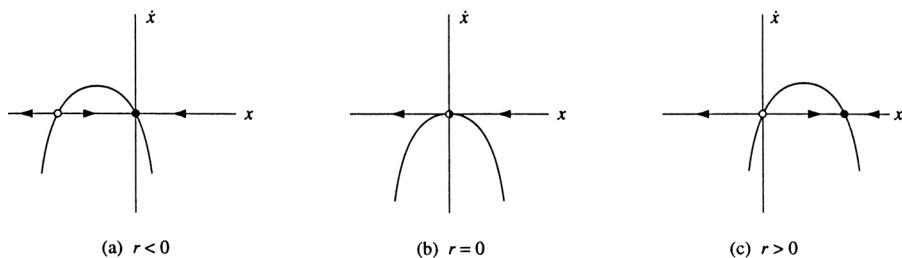


Figure 3.2.1

For $r < 0$, there is an unstable fixed point at $x^* = r$ and a stable fixed point at $x^* = 0$. As r increases, the unstable fixed point approaches the origin, and coalesces with it when $r = 0$. Finally, when $r > 0$, the origin has become unstable, and $x^* = r$ is now stable. Some people say that an **exchange of stabilities** has taken place between the two fixed points.

Please note the important difference between the saddle-node and transcritical bifurcations: in the transcritical case, the two fixed points don't disappear after the bifurcation—instead they just switch their stability.

Figure 3.2.2 shows the bifurcation diagram for the transcritical bifurcation. As in Figure 3.1.4, the parameter r is regarded as the independent variable, and the fixed points $x^* = 0$ and $x^* = r$ are shown as dependent variables.

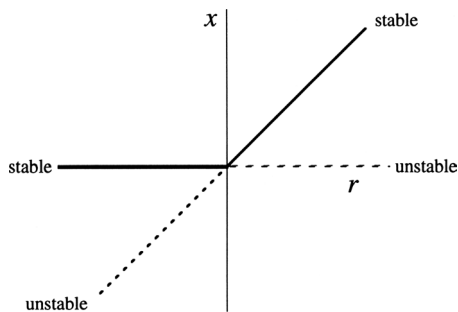


Figure 3.2.2

Example 3.2.1:

Show that the first-order system $\dot{x} = x(1 - x^2) - a(1 - e^{-bx})$ undergoes a transcritical bifurcation at $x = 0$ when the parameters a, b satisfy a certain equation, to be determined. (This equation defines a **bifurcation curve** in the (a, b) parameter space.) Then find an approximate formula for the fixed point that bifurcates from $x = 0$, assuming that the parameters are close to the bifurcation curve.

Solution: Note that $x = 0$ is a fixed point for all (a, b) . This makes it plausible that the fixed point will bifurcate transcritically, if it bifurcates at all. For small x , we find

$$\begin{aligned} 1 - e^{-bx} &= 1 - \left[1 - bx + \frac{1}{2}b^2x^2 + O(x^3)\right] \\ &= bx - \frac{1}{2}b^2x^2 + O(x^3) \end{aligned}$$

and so

$$\begin{aligned} \dot{x} &= x - a \left(bx - \frac{1}{2}b^2x^2\right) + O(x^3) \\ &= (1 - ab)x + \left(\frac{1}{2}ab^2\right)x^2 + O(x^3). \end{aligned}$$

Hence a transcritical bifurcation occurs when $ab = 1$; this is the equation for the bifurcation curve. The nonzero fixed point is given by the solution of $1 - ab + (\frac{1}{2}ab^2)x \approx 0$, i.e.,

$$x^* \approx \frac{2(ab - 1)}{ab^2}.$$

This formula is approximately correct only if x^* is small, since our series expansions are based on the assumption of small x . Thus the formula holds only when ab is close to 1, which means that the parameters must be close to the bifurcation curve. ■

Example 3.2.2:

Analyze the dynamics of $\dot{x} = r \ln x + x - 1$ near $x = 1$, and show that the system undergoes a transcritical bifurcation at a certain value of r . Then find new variables X and R such that the system reduces to the approximate normal form $\dot{X} \approx RX - X^2$ near the bifurcation.

Solution: First note that $x = 1$ is a fixed point for all values of r . Since we are interested in the dynamics near this fixed point, we introduce a new variable $u = x - 1$, where u is small. Then

$$\begin{aligned} \dot{u} &= \dot{x} \\ &= r \ln(1 + u) + u \\ &= r \left[u - \frac{1}{2}u^2 + O(u^3)\right] + u \\ &\approx (r + 1)u - \frac{1}{2}ru^2 + O(u^3). \end{aligned}$$

Hence a transcritical bifurcation occurs at $r_c = -1$.

To put this equation into normal form, we first need to get rid of the coefficient of u^2 . Let $u = av$, where a will be chosen later. Then the equation for v is

$$\dot{v} = (r + 1)v - \frac{1}{2}rav^2 + O(v^3).$$

So if we choose $a = 2/r$, the equation becomes

$$\dot{v} = (r + 1)v - v^2 + O(v^3).$$

Now if we let $R = r + 1$ and $X = v$, we have achieved the approximate normal form $\dot{X} \approx RX - X^2$, where cubic terms of order $O(X^3)$ have been neglected. In terms of the original variables, $X = v = u/a = \frac{1}{2}r(x - 1)$. ■

To be a bit more accurate, the theory of normal forms assures us that we can find a change of variables such that the system becomes $\dot{X} = RX - X^2$, with *strict*, rather than approximate, equality. Our solution above gives an approximation to the necessary change of variables. For careful treatments of normal form theory, see the books of Guckenheimer and Holmes (1983), Wiggins (1990), or Manneville (1990).

3.3 Laser Threshold

Now it's time to apply our mathematics to a scientific example. We analyze an extremely simplified model for a laser, following the treatment given by Haken (1983).

Physical Background

We are going to consider a particular type of laser known as a solid-state laser, which consists of a collection of special “laser-active” atoms embedded in a solid-state matrix, bounded by partially reflecting mirrors at either end. An external energy source is used to excite or “pump” the atoms out of their ground states (Figure 3.3.1).

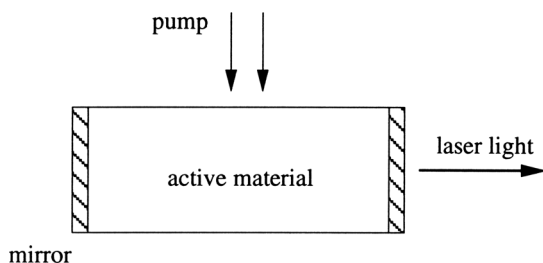


Figure 3.3.1

Each atom can be thought of as a little antenna radiating energy. When the pumping is relatively weak, the laser acts just like an ordinary lamp: the excited atoms oscillate independently of one another and emit randomly phased light waves.

Now suppose we increase the strength of the pumping. At first nothing different happens, but then suddenly, when the pump strength exceeds a certain threshold, the atoms begin to oscillate in phase—the lamp has turned into a **laser**. Now the trillions of little antennas act like one giant antenna and produce a beam of radiation that is much more coherent and intense than that produced below the laser threshold.

This sudden onset of coherence is amazing, considering that the atoms are being excited completely at random by the pump! Hence the process is *self-organizing*: the coherence develops because of a cooperative interaction among the atoms themselves.

Model

A proper explanation of the laser phenomenon would require us to delve into quantum mechanics. See Milonni and Eberly (1988) for an intuitive discussion.

Instead we consider a simplified model of the essential physics (Haken 1983, p. 127). The dynamical variable is the number of photons $n(t)$ in the laser field. Its rate of change is given by

$$\begin{aligned}\dot{n} &= \text{gain} - \text{loss} \\ &= GnN - kn.\end{aligned}$$

The gain term comes from the process of *stimulated emission*, in which photons stimulate excited atoms to emit additional photons. Because this process occurs via random encounters between photons and excited atoms, it occurs at a rate proportional to n and to the number of excited atoms, denoted by $N(t)$. The parameter $G > 0$ is known as the gain coefficient. The loss term models the escape of photons through the endfaces of the laser. The parameter $k > 0$ is a rate constant; its reciprocal $\tau = 1/k$ represents the typical lifetime of a photon in the laser.

Now comes the key physical idea: after an excited atom emits a photon, it drops down to a lower energy level and is no longer excited. Thus N decreases by the emission of photons. To capture this effect, we need to write an equation relating N to n . Suppose that in the absence of laser action, the pump keeps the number of excited atoms fixed at N_0 . Then the *actual* number of excited atoms will be reduced by the laser process. Specifically, we assume

$$N(t) = N_0 - \alpha n,$$

where $\alpha > 0$ is the rate at which atoms drop back to their ground states. Then

$$\begin{aligned}\dot{n} &= Gn(N_0 - \alpha n) - kn \\ &= (GN_0 - k)n - (\alpha G)n^2.\end{aligned}$$

We're finally on familiar ground—this is a first-order system for $n(t)$. [Figure 3.3.2](#) shows the corresponding vector field for different values of the pump strength N_0 . Note that only positive values of n are physically meaningful.

When $N_0 < k/G$, the fixed point at $n^* = 0$ is stable. This means that there is no stimulated emission and the laser acts like a lamp. As the pump strength N_0 is increased, the system undergoes a transcritical bifurcation when $N_0 = k/G$. For $N_0 > k/G$, the origin loses stability and a stable fixed point appears at $n^* = (GN_0 - k)/\alpha G > 0$, corresponding to spontaneous laser

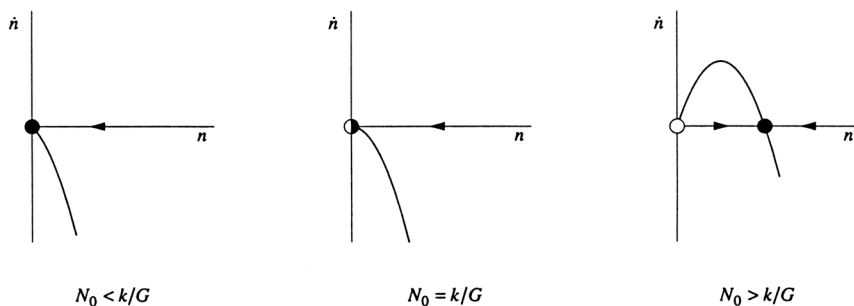


Figure 3.3.2

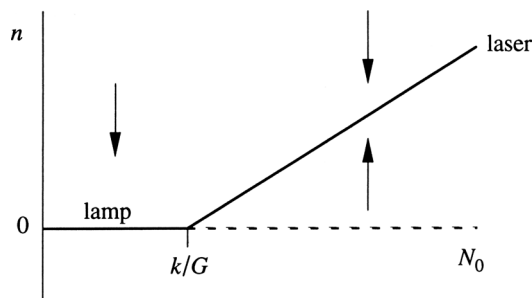


Figure 3.3.3

action. Thus $N_0 = k/G$ can be interpreted as the *laser threshold* in this model. Figure 3.3.3 summarizes our results.

Although this model correctly predicts the existence of a threshold, it ignores the dynamics of the excited atoms, the existence of spontaneous emission, and several other complications. See Exercises 3.3.1 and 3.3.2 for improved models.

3.4 Pitchfork Bifurcation

We turn now to a third kind of bifurcation, the so-called pitchfork bifurcation. This bifurcation is common in physical problems that have a *symmetry*. For example, many problems have a spatial symmetry between left and right. In such cases, fixed points tend to appear and disappear in symmetrical pairs. In the buckling example of Figure 3.0.1, the beam is stable in the vertical position if the load is small. In this case there is a stable fixed point corresponding to zero deflection. But if the load exceeds the buckling threshold, the beam may buckle to *either* the left or the right. The vertical position has gone unstable,

and two new symmetrical fixed points, corresponding to left- and right-buckled configurations, have been born.

There are two very different types of pitchfork bifurcation. The simpler type is called *supercritical*, and will be discussed first.

Supercritical Pitchfork Bifurcation

The normal form of the supercritical pitchfork bifurcation is

$$\dot{x} = rx - x^3. \quad (1)$$

Note that this equation is *invariant* under the change of variables $x \rightarrow -x$. That is, if we replace x by $-x$ and then cancel the resulting minus signs on both sides of the equation, we get (1) back again. This invariance is the mathematical expression of the left-right symmetry mentioned earlier. (More technically, one says that the vector field is *equivariant*, but we'll use the more familiar language.)

Figure 3.4.1 shows the vector field for different values of r .

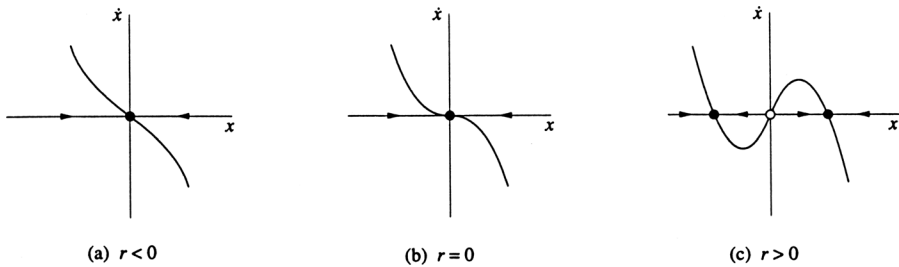


Figure 3.4.1

When $r < 0$, the origin is the only fixed point, and it is stable. When $r = 0$, the origin is still stable, but much more weakly so, since the linearization vanishes. Now solutions no longer decay exponentially fast—instead the decay is a much slower algebraic function of time (recall Exercise 2.4.9). This lethargic decay is called *critical slowing down*. Finally, when $r > 0$, the origin has become unstable. Two new stable fixed points appear on either side of the origin, symmetrically located at $x^* = \pm\sqrt{r}$.

The reason for the term “pitchfork” becomes clear when we plot the bifurcation diagram (Figure 3.4.2). Actually, pitchfork trifurcation might be a better word!

Example 3.4.1:

Equations similar to $\dot{x} = -x + \beta \tanh x$ arise in statistical mechanical models of magnets and neural networks (see Exercise 3.6.7 and Palmer 1989). Show that this equation undergoes a supercritical pitchfork bifurcation as β is varied. Then give a *numerically accurate* plot of the fixed points for each β .

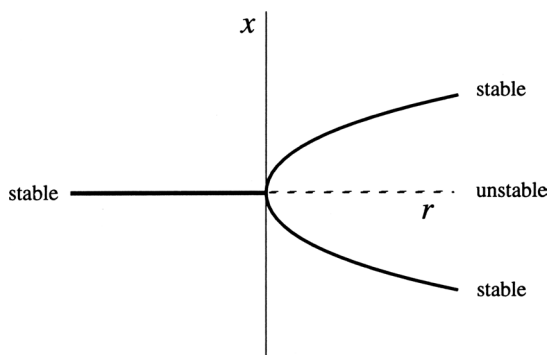


Figure 3.4.2

Solution: We use the strategy of [Example 3.1.2](#) to find the fixed points. The graphs of $y = x$ and $y = \beta \tanh x$ are shown in [Figure 3.4.3](#); their intersections correspond to fixed points.

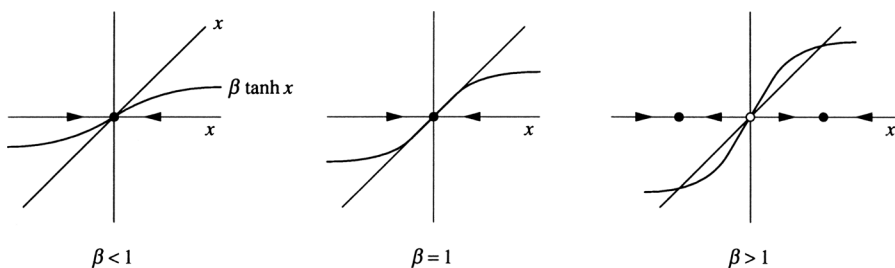


Figure 3.4.3

The key thing to realize is that as β increases, the $\tanh$ curve becomes steeper at the origin (its slope there is β). Hence for $\beta < 1$ the origin is the only fixed point. A pitchfork bifurcation occurs at $\beta = 1$, $x^* = 0$, when the $\tanh$ curve develops a slope of 1 at the origin. Finally, when $\beta > 1$, two new stable fixed points appear, and the origin becomes unstable.

Now we want to compute the fixed points x^* for each β . Of course, one fixed point always occurs at $x^* = 0$; we are looking for the other, nontrivial fixed points. One approach is to solve the equation $x^* = \beta \tanh x^*$ numerically, using the Newton–Raphson method or some other root-finding scheme. (See Press et al. (2007) for a friendly and informative discussion of numerical methods.)

But there’s an easier way, which comes from changing our point of view. Instead of studying the dependence of x^* on β , we think of x^* as the *independent* variable, and then compute $\beta = x^* / \tanh x^*$. This gives us a table of pairs (x^*, β) . For each pair, we plot β horizontally and x^* vertically. This yields the bifurcation diagram ([Figure 3.4.4](#)).